SNC Standard Data Format

What is a SNC file?

A SNC (S-N Curve) file corresponds to the output of the S-N Curve module.
It contains the fitted fatigue curve data for one or several stress ratios and is composed of two files:

- one JSON file containing method parameters and metadata
- one CSV file containing the curve data points

File naming convention

File:

- SNC_{ResearcherLastname}_{YYYY-MM}_{axis}_{methods}.json
- SNC_{ResearcherLastname}_{YYYY-MM}_{axis}_{methods}.csv

with:

- YYYY-MM = starting month of the experiment
- axis = test direction or loading case, depending on the method or material convention
  - X = longitudinal
  - Y = transverse
  - F = under shear loading
- methods = optional list of methods used in the different modules

File format requirements

- Encoding: UTF-8
- Separator: comma (,)
- File type: .csv and .json
- Header required
- Units: MPa

Required columns for CSV (column names must be exact)

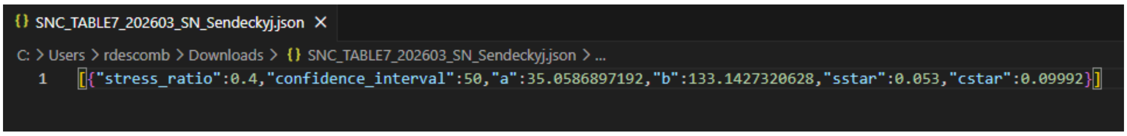
Variable name	Description	Symbol	Unit	Data type	Mandatory
stress_ratio	Stress ratio	R	[-]	double	y
cycles_to_failure	Number of cycles to failure	N	[-]	int	y
stress_max	Max stress	sigma_max	[MPa]	double	y
stress_lowerbound	Stress at failure (lower bound)	sigma_max	[MPa]	double	
stress_upperbound	Stress at failure (upper bound)	sigma_max	[MPa]	double	

stress_lowerbound and stress_upperbound correspond to the lower and upper bounds of the fitted S-N curve, typically representing the confidence interval of the model.

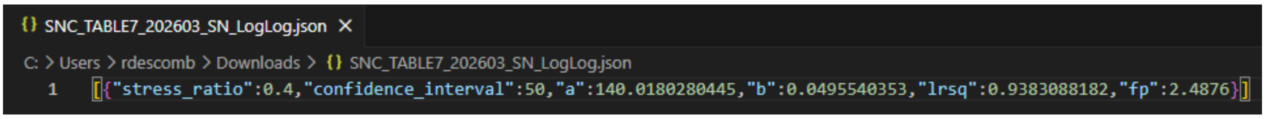
JSON structure

```
[
        {
                "stress_ratio": double, // Stress ratio (R) [-]
                "confidence_interval": double, // confidence bounds, usually 5, 95% (rsql) [-]
                "a": double, // S-N Curve parameter [-]
                "b": double, // S-N Curve parameter [1/MPa]
                "lrsq": double, // coefficient of determination of the regression (R²), optional
                "fp": double, // linearity criterion used during fitting, optional        },
        ...
]
```

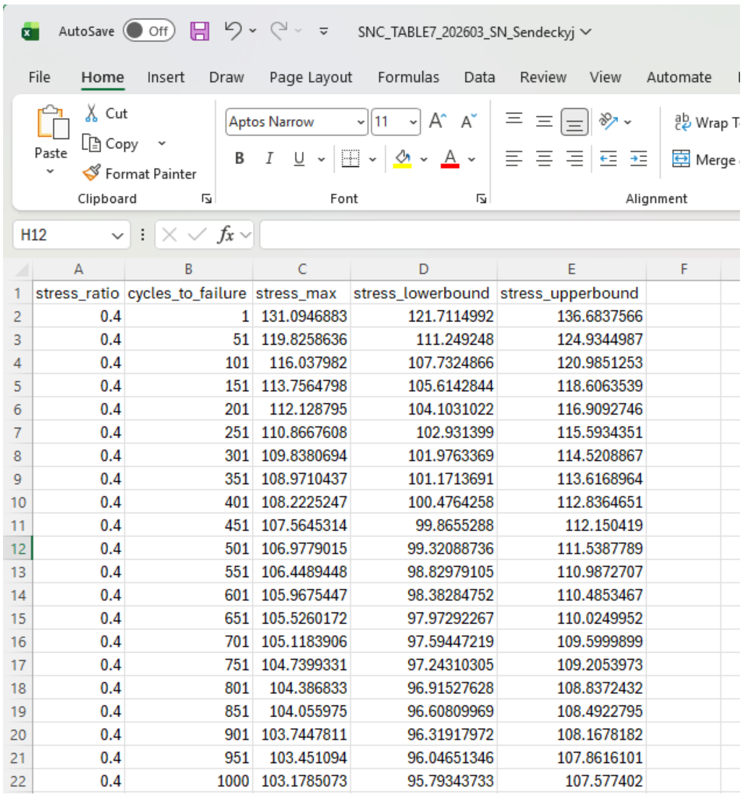
Example of valid files



JSON file (Sendeckyj)



JSON file (LogLog)



stress_ratio	cycles_to_failure	stress_max	stress_lowerbound	stress_upperbound
0.4	1	131.0946883	121.7114992	136.6837566
0.4	51	119.8258636	111.249248	124.9344987
0.4	101	116.037982	107.7324866	120.9851253
0.4	151	113.7564798	105.6142844	118.6063539
0.4	201	112.128795	104.1031022	116.9092746
0.4	251	110.8667608	102.931399	115.5934351
0.4	301	109.8380694	101.9763369	114.5208867
0.4	351	108.9710437	101.1713691	113.6168964
0.4	401	108.2225247	100.4764258	112.8364651
0.4	451	107.5645314	99.8655288	112.150419
0.4	501	106.9779015	99.32088736	111.5387789
0.4	551	106.4489448	98.82979105	110.9872707
0.4	601	105.9675447	98.38284752	110.4853467
0.4	651	105.5260172	97.97292267	110.0249952
0.4	701	105.1183906	97.59447219	109.5999899
0.4	751	104.7399331	97.24310305	109.2053973
0.4	801	104.386833	96.91527628	108.8372432
0.4	851	104.055975	96.60809969	108.4922795
0.4	901	103.7447811	96.31917972	108.1678182
0.4	951	103.451094	96.04651346	107.8616101
0.4	1000	103.1785073	95.79343733	107.577402

LDS file

The JSON file provides the fitted model parameters (i.e. the equation definition), while the CSV file contains the computed S–N values that can be used as input for the CLD calculation.

For the CSV file, Microsoft Excel works well for editing and saving the data.
However, other tools such as LibreOffice Calc, Google Sheets, or any text editor (e.g., Notepad++, VS Code) can also be used, provided the file is saved in CSV format (UTF-8 encoding).